

4 CRC calculation unit

This section applies to the whole STM32F4xx family, unless otherwise specified.

4.1 CRC introduction

The CRC (cyclic redundancy check) calculation unit is used to get a CRC code from a 32-bit data word and a fixed generator polynomial.

Among other applications, CRC-based techniques are used to verify data transmission or storage integrity. In the scope of the EN/IEC 60335-1 standard, they offer a means of verifying the flash memory integrity. The CRC calculation unit helps compute a signature of the software during runtime, to be compared with a reference signature generated at link-time and stored at a given memory location.

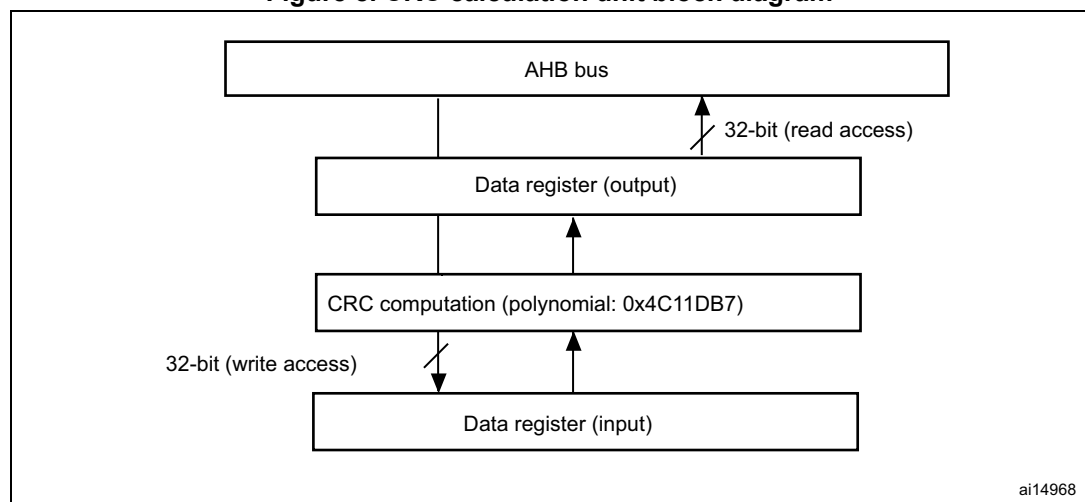
4.2 CRC main features

- Uses CRC-32 (Ethernet) polynomial: 0x4C11DB7

$$X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$$
- Single input/output 32-bit data register
- CRC computation done in 4 AHB clock cycles (HCLK)
- General-purpose 8-bit register (can be used for temporary storage)

The block diagram is shown in [Figure 8](#).

Figure 8. CRC calculation unit block diagram



4.3 CRC functional description

The CRC calculation unit mainly consists of a single 32-bit data register, which:

- is used as an input register to enter new data in the CRC calculator (when writing into the register)
- holds the result of the previous CRC calculation (when reading the register)

Each write operation into the data register creates a combination of the previous CRC value and the new one (CRC computation is done on the whole 32-bit data word, and not byte per byte).

The write operation is stalled until the end of the CRC computation, thus allowing back-to-back write accesses or consecutive write and read accesses.

The CRC calculator can be reset to 0xFFFF FFFF with the RESET control bit in the CRC_CR register. This operation does not affect the contents of the CRC_IDR register.

4.4 CRC registers

The CRC calculation unit contains two data registers and a control register. The peripheral The CRC registers have to be accessed by words (32 bits).

4.4.1 Data register (CRC_DR)

Address offset: 0x00

Reset value: 0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DR [31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DR [15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **Data register bits**

Used as an input register when writing new data into the CRC calculator.

Holds the previous CRC calculation result when it is read.

4.4.2 Independent data register (CRC_IDR)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								IDR[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **General-purpose 8-bit data register bits**

Can be used as a temporary storage location for one byte.

This register is not affected by CRC resets generated by the RESET bit in the CRC_CR register.

4.4.3 Control register (CRC_CR)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														RESET	
														W	

Bits 31:1 Reserved, must be kept at reset value.

Bit 0 **RESET bit**

Resets the CRC calculation unit and sets the data register to 0xFFFF FFFF.

This bit can only be set, it is automatically cleared by hardware.

4.4.4 CRC register map

The following table provides the CRC register map and reset values.

Table 22. CRC calculation unit register map and reset values

Offset	Register	31-24	23-16	15-8	7	6	5	4	3	2	1	0
0x00	CRC_DR	Data register										
	Reset value	0xFFFF FFFF										
0x04	CRC_IDR	Reserved			Independent data register							
	Reset value				0x00							
0x08	CRC_CR	Reserved										RESET
	Reset value											0